Decomposition Using Multivalued Attribute (4NF) DBMS

> **Definition:** A relation is in **Fourth Normal Form (4NF)** if it is in **BCNF** and for every non-trivial **Multivalued Dependency (MVD) X ->-> Y**, **X is a Super Key**.

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1. Detailed Technical Explanation

Multivalued Dependency (MVD):

A multivalued dependency **X woheadrightarrow Y** (read as "X multidetermines Y") exists in schema R if the presence of a pair of tuples **(t_1, t_2)** with **t_1[X] = t_2[X]** implies that there must also exist tuples **t_3** and **t_4** combining **X**, **Y**, and the remaining attributes **Z = R - (X \cup Y)**.

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MVD S

### Example: 4NF Decomposition

#### Relation STUDENT_INFO(Student_ID, Mobile_No, Skill)

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#### Data Tuple Redundancy (UNF/BCNF Violation):

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*(Note: Adding 1 new skill for student 101 requires inserting 2 new rows because Mobile_No and Skill are independent!)*

#### 4NF Decomposition Algorithm:

If **X

#### Decomposed 4NF Tables:

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## 2. Core Concepts & Memory Keywords

- **Multivalued Dependency (MVD):** Independence between two multi-valued attributes associated with the same key.

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3. Must-Write Points for Exams

- 4NF handles independent multi-valued attributes that BCNF cannot resolve.

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4. Quick Recall Flow

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BCNF
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